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Source selected

Scenes: SEMF - Wait, SEMF - Talker, SEMF - Stream, SEMF - Agoria Stream, HALL

Scenes: Spatiality, Architecture, Web, Logo (mini), Room, Theater, Studio, Waves

Audio Mixer: Custom Audio (4.4-48), MIDI (1.0-48), Music (1.0-48), Voice (1.0-48)

Scene Transitions: Fade, Duration: 200ms

Controls: Clip Blending, Clip Recording, Start virtual camera, Scale Mode, Settings, Exit

15:01

# welcome

INFORMATION

- # announcements
- # style
- # workshop
- # questions
- # suggestions
- # feedback

AGORA

- # agenda
- # introductions
- # events
- # agenda

SPACIOUS SPATIALITY 2022

- # general\_discussion
- # art\_repository
- # conceptual\_spaces
- # spatial\_thinking
- # biological\_spaces
- # everyday\_art
- # geometric\_computation
- # emergent\_spaces
- # geometry\_of\_music
- # time\_and\_spatiality
- # meaning\_spaces
- # challenges\_of\_space

RECOMMENDATIONS

- # reading
- # listening
- # watching
- # playing
- # attending

SEMFLQQUIN

- # art\_model
- # phys\_math

Zephyr

# Welcome to #general\_disc

This is the start of the #general\_discussion channel.

23 Nov 2022

Please join us for our opening session via Zoom!

In case you join late / miss it, the session will be live-streamed

<https://www.zoom.us/j/97558982046>

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Opening Session



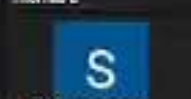


Recording...

Now you are in a meeting. Please click on the 'Join' button.

View Options

Gardiner's Peter has started screen sharing

|                                                                                                             |                                                                                                             |                                                                                                          |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <br>Mandy Barrer         | <br>Carlos Zapata-Correa | <br>Gardiner's Peter  |
| <br>Alvaro Moreno Valler | <br>Daniel Cima          | barbaratovsky<br>Barbara Tovsky                                                                          |
| Micah F.<br>Micah F.                                                                                        | <br>Francisco Gallego    | Binoy<br>Binoy                                                                                           |
| Thomas B.<br>Thomas B.                                                                                      | <br>Wida Altman          | <br>Maximilian Schich |
| <br>Shahenda Jaffer      | Martin Green<br>Martin Green                                                                                | Ben MacVicar<br>Ben MacVicar                                                                             |
| Tilman Ohm<br>Tilman Ohm                                                                                    | <br>Kathleen Overmann   | <br>Taja Kozak       |
| <br>U. W.              | Juan Malero<br>Juan Malero                                                                                  | Kella Moral<br>Kella Moral                                                                               |
| <br>Chris Mitchell     | Mike Tamm<br>Mike Tamm                                                                                      |                                                                                                          |

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+ Dela

Klistra in
Ny bild
Konvertera till SmartArt
Bild
Ritning



Peter Gärdenfors

### Conceptual spaces and the geometry of word meanings

#### 1st paradigm: Symbolism

The computer as a metaphor for cognition

- The brain functions as a Turing machine
- The brain is an information-processing system
- The brain has a central processor and memory units
- The brain works on the basis of a (symbolic) code by which all information is represented



#### 2nd paradigm: Connectionism

Cognitive processes can be modelled in artificial neural networks

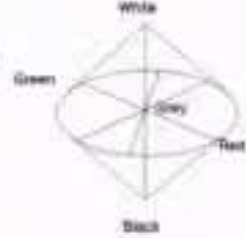


Active perception: Cognitive processes can be identified with the input and output activities of the neurons in the brain

#### 3rd paradigm: Spatial models

Cognition can be modelled in topological and geometrical structures

- Conceptual spaces
- Information is organized in spatial structures
- Distances represent inverse similarity
- Connections to spatial representations in the hippocampal complex

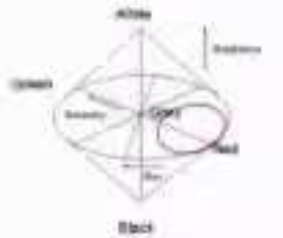


#### Conceptual spaces

- Consists of a number of dimensions
- Dimensions are sorted into domains
- Space, size, colour, weight, force, temperature, shape, ...
- Domains have topological or geometric structures
- Concepts are represented as convex regions of domains



#### The color spindle

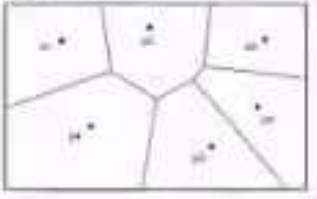


#### Why convexity?

- Connects to prototype theory
- A robin is more typical bird than an ostrich
- Children learn prototypical meanings first

#### Categorization in conceptual spaces

Nearest triangulation around prototype objects divides conceptual spaces into categories based on the nearest neighbour rule, i.e., each object is associated with the prototype closest to it.



#### Why convexity?

- Connects to prototype theory
- Handles vague concepts

#### Learning from few examples



#### Learning from few examples



#### Why convexity?

- Connects to prototype theory
- Handles vague concepts

Bild 1 av 45 Svenska (Finland) Tillgänglighet: Utlämnad
100%

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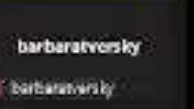
  
Maddy Beller

  
Carlos Zapata-Correa

  
Gardar Peters

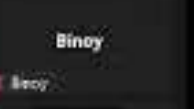
  
barbaratvovsky

  
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Mikah F.

  
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Thomas B

  
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Shahenaz Jaffer

  
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Tilman Ohm

  
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Juan Moleiro

  
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Mike Tamme

  
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# Conceptual spaces and the geometry of word meanings



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# 1st paradigm: Symbolism

## The computer as a metaphor for cognition





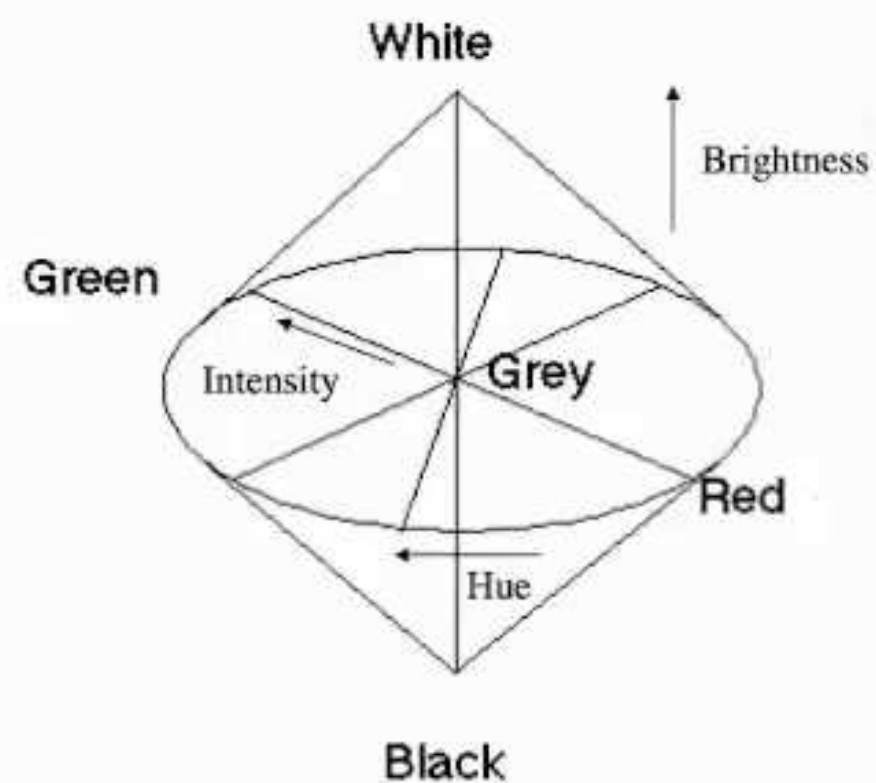
## 2nd paradigm: Connectionism

Cognitive processes can be modelled in artificial neural networks





# The color spindle



# Word meanings have geometric structures



Valentin A. Bazh...

all candidates page



Humans are excellent at identifying actions



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# Representational hypothesis for actions

- The fundamental cognitive representation of an action is the *pattern of forces* that generates it
- Actions are more or less *similar* and show prototype effects
- An *action concept* is a convex region in the space of force patterns



Malt et al. 2014, J. of Memory and Language







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# Conceptual spaces and the geometry of word meanings

